

No. of Functional Features Included in the Solution

Date	1 july 2026
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine”
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation Engine	Analyzes soil parameters (N, P, K, pH) and environmental factors to predict optimal crops	Backend/ML	DONE	
2	Gemini AI integration	Generates context-aware, data-driven agricultural insights and natural language responses.	API integration	DONE	
3	Intractive dashboard	Web interface allowing users to easily input soil data and view AI recommendations.	User Interface	DONE	
4	RESTfullAPI	EndpointsConnects the client-side frontend to the backend server to process agricultural data	Backend/API	DONE	
5	Local Data Management	Persists application metadata and handles local data storage	Database /storage	DONE	
6	Secure Environment config	Secures sensitive API credentials utilizing hidden environment variables.	security	DONE	
7	Source Control Tracking	Manages code versioning and online remote backup via GitHub	deployment	DONE	
8					

Feature Summary:

Metric	Count / Value
Total Features Planned	7
Total Features Implemented	7
Core / Must-Have Features	4

Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	User interface (UI)	1
2	Backend / Logic	Backend/Logic	1
3	Database / Storage	Database/Storage	1
4	API / Integration	API/Integration	2
5	Security / Authentication	Security/Authentication	1